```
import pandas as pd

df = pd.read_excel('/content/data set (1).xlsx')

df.head()
```

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	
0	750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	7	2
1	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	
2	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	7	1
3	123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	8	2
4	373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	7	3

Next steps: Generate code with df | New interactive sheet

```
import sqlite3

conn = sqlite3.connect('sales_database.db')

df.to_sql('sales', conn, if_exists='replace', index=False)

print("Sales table created successfully")
```

```
Sales table created successfully
```

```
cursor = conn.cursor()

cursor.execute("PRAGMA table_info(sales)")

cursor.fetchall()
```

```
[(0, 'Invoice ID', 'TEXT', 0, None, 0),
 (1, 'Branch', 'TEXT', 0, None, 0),
 (2, 'City', 'TEXT', 0, None, 0),
 (3, 'Customer type', 'TEXT', 0, None, 0),
 (4, 'Gender', 'TEXT', 0, None, 0),
 (5, 'Product line', 'TEXT', 0, None, 0),
```

```
(6, 'Unit price', 'REAL', 0, None, 0),
(7, 'Quantity', 'INTEGER', 0, None, 0),
(8, 'Tax 5%', 'REAL', 0, None, 0),
(9, 'Total', 'REAL', 0, None, 0),
(10, 'Date', 'TEXT', 0, None, 0),
(11, 'Time', 'TIME', 0, None, 0),
(12, 'Payment', 'TEXT', 0, None, 0),
(13, 'cogs', 'REAL', 0, None, 0),
(14, 'gross margin percentage', 'REAL', 0, None, 0),
(15, 'gross income', 'REAL', 0, None, 0),
(16, 'Rating', 'REAL', 0, None, 0)]
```

```
cursor.execute("SELECT * FROM sales")

cursor.fetchall()
```

```
 'Yangon',
 'Member',
 'Female',
 'Fashion accessories',
 88.34,
 7,
 30.919,
 649.299,
 '2/18/2019',
 '13:28:00.000000',
 'Cash',
 618.38,
 4.761904762,
 30.919,
 6.6)]
```

```
print(df.columns.tolist())
```

```
['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender', 'Product line', '
```

```
cursor.execute("SELECT * FROM sales ORDER BY 1 ASC")

cursor.fetchall()
```

```
 '1/26/2019',
  '12:43:00.000000',
  'Cash',
  21.32,
  4.761904762,
  1.066,
  5.9),
 ('898-04-2717',
  'A',
  'Yangon',
  'Normal',
  'Male',
  'Fashion accessories',
  76.4,
  9,
  34.38,
  721.98,
  '3/19/2019',
  '15:49:00.000000',
  'Ewallet',
  687.6,
  4.761904762,
  34.38,
  7.5)]
```

```
print(df.columns.tolist())
```

```
['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender', 'Product line', '
```

```
cursor.execute("""
CREATE TABLE customers (
    Customer_ID TEXT,
    Customer_Name TEXT
)
""")

print("Customers table created successfully")
```

```
Customers table created successfully
```

```
print(df.columns.tolist())
```

```
['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender', 'Product line', '
```

```
print(df.columns.tolist())
```

```
['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender', 'Product line', '
```

```
cursor.execute("PRAGMA table_info(customers)")
print(cursor.fetchall())
```

```
[(0, 'Customer_ID', 'TEXT', 0, None, 0), (1, 'Customer_Name', 'TEXT', 0, None
```

```
print(df.columns.tolist())
```

```
['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender', 'Product line', '
```

```
branch_df = df[['Branch', 'City']].drop_duplicates()

branch_df.to_sql('branch_info', conn, if_exists='replace', index=False)

print("Branch table created successfully")
```

```
Branch table created successfully
```

```
cursor.execute("""
SELECT *
FROM sales
LEFT JOIN branch_info
ON sales.Branch = branch_info.Branch
""")

cursor.fetchall()
```

```
 3.291,
 4.1,
 'A',
 'Yangon'),
('849-09-3807',
 'A',
 'Yangon',
 'Member',
 'Female',
 'Fashion accessories',
 88.34,
 7,
 30.919,
 649.299,
 '2/18/2019',
 '13:28:00.000000',
 'Cash',
 618.38,
 4.761904762,
 30.919,
 6.6,
 'A',
 'Yangon')]
```

```
cursor.execute("""
SELECT *
FROM branch_info
RIGHT JOIN sales
ON branch_info.Branch = sales.Branch
""")

cursor.fetchall()
```

```
 'Fashion accessories',
 17.49,
 10,
 8.745,
 183.645,
 '2/22/2019',
 '18:35:00.000000',
 'Ewallet',
 174.9,
 4.761904762,
 8.745,
 6.6),
('B',
 'Mandalay',
 '303-96-2227',
 'B',
 'Mandalay',
 'Normal',
 'Female',
 'Home and lifestyle',
 97.38,
 10,
 48.69,
 1022.49,
 '2019-02-03 00:00:00',
 '17:16:00.000000',
 'Ewallet',
 973.8,
 4.761904762,
 48.69,
 4.4)]
```

```
cursor.execute("""
SELECT *
FROM sales
INNER JOIN branch_info
ON sales.Branch = branch_info.Branch
""")

cursor.fetchall()
```

```
  'A',
  'Yangon'),
 ('347-56-2442',
  'A',
  'Yangon',
  'Normal',
  'Male',
  'Home and lifestyle',
  65.82,
  1,
  3.291,
  69.111,
  '2/22/2019',
  '15:33:00.000000',
  'Cash',
  65.82,
  4.761904762,
  3.291,
  4.1,
  'A',
  'Yangon'),
 ('849-09-3807',
  'A',
  'Yangon',
  'Member',
  'Female',
  'Fashion accessories',
  88.34,
  7,
  30.919,
  649.299,
  '2/18/2019',
  '13:28:00.000000',
  'Cash',
  618.38,
  4.761904762,
  30.919,
  6.6,
  'A',
  'Yangon')]
```

```
cursor.execute("""
SELECT *
FROM sales
WHERE `Total` > (
    SELECT AVG(`Total`)
    FROM sales
)
""")

cursor.fetchall()
```

```
 ('602-16-6955',
  'B',
  'Mandalay',
  'Normal',
  'Female',
  'Sports and travel',
  76.6,
  10,
  38.3,
  804.3,
  '1/24/2019',
  '18:10:00.000000',
  'Ewallet',
  766.0,
  4.761904762,
  38.3,
  6.0),
 ('303-96-2227',
  'B',
  'Mandalay',
  'Normal',
  'Female',
  'Home and lifestyle',
  97.38,
  10,
  48.69,
  1022.49,
  '2019-02-03 00:00:00',
  '17:16:00.000000',
  'Ewallet',
  973.8,
  4.761904762,
  48.69,
  4.4),
 ('849-09-3807',
  'A',
  'Yangon',
  'Member',
  'Female',
  'Fashion accessories',
  88.34,
  7,
  30.919,
  649.299,
  '2/18/2019',
  '13:28:00.000000',
  'Cash',
  618.38,
  4.761904762,
  30.919,
  6.6)]
```

```
cursor.execute("""
SELECT SUM(Total)
FROM sales
""")

print(cursor.fetchall())
```

```
[(322966.749,)]
```

```
cursor.execute("""
SELECT AVG(Total)
FROM sales
""")

print(cursor.fetchall())
```

```
[(322.966749,)]
```

```
cursor.execute("""
CREATE VIEW sales_analysis AS
SELECT *
FROM sales
""")

print("View created successfully")
```

```
View created successfully
```

```
cursor.execute("""
SELECT * FROM sales_analysis
""")

cursor.fetchall()
```

```
 'Male',
 'Home and lifestyle',
 65.82,
 1,
 3.291,
 69.111,
 '2/22/2019',
 '15:33:00.000000',
 'Cash',
 65.82,
 4.761904762,
 3.291,
 4.1),
('849-09-3807',
 'A',
 'Yangon',
 'Member',
 'Female',
 'Fashion accessories',
 88.34,
 7,
 30.919,
 649.299,
 '2/18/2019',
 '13:28:00.000000',
 'Cash',
 618.38,
 4.761904762,
 30.919,
```

```
cursor.execute("""
CREATE INDEX idx_sales_branch
ON sales(Branch)
""")

print("Index created successfully")
```

```
Index created successfully
```